

The Business Impact of MaiaEdge for Cloud On-Ramp, Wholesale Connectivity, and Ethernet over DIA

Executive Summary

Enterprise demand for private cloud connectivity continues to accelerate as organizations increasingly rely on cloud-based applications, AI workloads, and hybrid multi-cloud architectures. Regional service providers and colocation providers are under growing pressure to deliver secure, high-performance cloud connectivity services rapidly and cost-effectively. However, many providers are missing revenue opportunities because they cannot deliver cloud connectivity and access to locations outside their footprint. This is complicated by the fact that cloud on-ramps and partner connectivity require manual processes which are resource-intensive and slow to deliver.

Regional service providers and colocation providers face two distinct challenges:

1. Enterprises demand cloud connectivity, but not all providers can deliver it; the ones that do deal with manual and costly provisioning.
2. Interconnecting with partner providers is labor intensive plus providers lose visibility and control once traffic leaves their network.

Solution: MaiaEdge provides a carrier-neutral interconnection engine, automated private path provisioning across any underlying transport, and simplified cloud on-ramp connectivity through API integrations with Equinix and Megaport.

The economic benefits of the MaiaEdge solution are:

- Expanding network reach through partners.
- Lowering the barrier to entry for cloud on-ramp services.
- Monetizing existing cloud connectivity through wholesale revenue opportunities
- Reducing the cost of private Layer 2 service delivery over DIA.

ACG Research developed three economic models showing MaiaEdge business impacts:

1. Cloud on-ramp delivery TCO.
2. Wholesale cloud connectivity business case.
3. Ethernet over DIA delivery TCO.

The results of the ACG business case analysis showed:

- A cloud on-ramp delivery model using MaiaEdge resulted in a **67% reduction in five-year TCO**.
- Operators using MaiaEdge to wholesale cloud connectivity can generate **margins approaching 67% with payback in approximately six months**.
- Ethernet over DIA with MaiaEdge resulted in a **53% TCO reduction** over dedicated L2 service.

Overall, the analysis demonstrates that MaiaEdge is both a cost optimization and revenue growth platform that enables service providers to deliver scalable, automated, and economically efficient cloud interconnection services.

MaiaEdge Solution

The MaiaEdge solution enables automated private path provisioning, cloud on-ramp monetization, partner reach expansion, and visibility across provider domains.

MaiaEdge combines a 1RU Path Border Controller (PBC) deployed at key interconnection points with a cloud-native Path Computation Engine (PCE) that automates private Ethernet path provisioning over any transport, enforces traffic engineering and SLA policy, and provides end-to-end visibility across both owned and partner networks. Together, these components allow operators to launch cloud on-ramp services, sell more capacity across existing cloud connections, offer wholesale cloud connectivity to partners, and deliver private Ethernet services over available transport options where appropriate.

Economic and TCO Benefits of MaiaEdge

MaiaEdge delivers significant economic benefits across many use cases. This paper presents the business case for three use cases:

1. Delivering and monetizing cloud on-ramp services.
2. Selling wholesale access to existing cloud on-ramp services.
3. Establishing Ethernet over DIA with MaiaEdge vs. traditional Carrier Ethernet.

This paper compares the Present Mode of Operation (PMO) with the MaiaEdge Future Mode of Operation (FMO) and quantifies the financial and operational advantages of the MaiaEdge solution.

Cloud On-Ramp Use Case

As enterprise reliance on AWS, Azure, Google Cloud, and Oracle Cloud grows, regional service providers and colocation providers face increasing pressure to deliver secure, high-performance cloud on-ramp services quickly and cost-effectively.

Cloud On-Ramp Present Mode of Operation (PMO)

To provide cloud access today, service providers and colocation operators typically connect to cloud on-ramps through fabrics such as Equinix or Megaport. Each customer connection often requires custom provisioning, NNI coordination, and routing infrastructure or cloud-router services to support multi-cloud connectivity. These manual steps add cost, delay service activation, and make cloud on-ramp services harder to scale.

Providers also tend to use only about 30% of available cloud exchange port capacity because they lack the traffic engineering and SLA-aware optimization needed to use shared cloud capacity efficiently.

Future Mode of Operation (FMO) with MaiaEdge

The MaiaEdge architecture automates and simplifies cloud on-ramp provisioning while significantly improving infrastructure efficiency and operational scalability. In the MaiaEdge solution, enterprise access circuits including DIA, MPLS, and Carrier Ethernet are dynamically mapped to cloud on-ramp virtual circuits by the MaiaEdge PBC using automated orchestration and traffic engineering. The PCE continuously monitors real-time traffic metrics across a shared Equinix or Megaport port, so that when traffic bursts occur, the PCE dynamically load balances connections.

This capability enables providers to:

- Increase utilization of expensive cloud exchange ports.
- Support customer bursting above committed information rates (CIR).
- Dynamically prioritize traffic flows.
- Optimize ingress and egress traffic engineering.
- Reduce stranded bandwidth capacity.
- Lower overall cost per delivered gigabit.

For example, a provider may offer a customer a 1 Gbps CIR service capable of bursting to 5 Gbps while still maintaining SLA performance through intelligent traffic engineering and dynamic path optimization.

TCO Model for Cloud On-Ramp Services

To quantify the operational and economic benefits of MaiaEdge, ACG Research developed a five-year Total Cost of Ownership (TCO) model comparing the PMO with a MaiaEdge-based FMO cloud on-ramp architecture. The analysis period spans January 2026 through December 2030 and evaluates operational expenses (OpEx), network infrastructure costs, provisioning labor, and operational staffing requirements.

TCO Model Assumptions

For both PMO and FMO scenarios, the model assumes:

- 1,000 enterprise customer sites
- 5% annual site growth
- 500 Mbps average bandwidth per site
- 15% annual traffic growth
- Five-year TCO analysis period (2026–2030)
- Connectivity through 100G Equinix cloud exchange ports

The model also assumes a five-year MaiaEdge PBC license.

Cloud Connectivity Cost Assumptions

The following average recurring costs were used in the model:

- Equinix 100G Port: \$1,500 monthly recurring charge (MRC)
- Equinix 500 Mbps virtual circuit: \$150 MRC
- Cross-connects between the service provider and Equinix: \$500 MRC per port

These costs were applied consistently across the PMO and FMO scenarios to isolate the operational and utilization benefits enabled by MaiaEdge.

PMO Assumptions

The PMO model represents a traditional cloud on-ramp architecture using manual provisioning and limited traffic engineering capabilities.

Key assumptions include:

- 30% average cloud exchange port utilization
- Two hours of provisioning labor per customer site
- One hour of annual network operations labor per customer site

The low utilization assumption reflects the operational limitations of traditional environments where providers typically avoid aggressive oversubscription because they lack SLA-aware traffic engineering and dynamic bandwidth management capabilities.

FMO Assumptions

The MaiaEdge Future Mode of Operation (FMO) model assumes automated provisioning, advanced traffic engineering, and intelligent bandwidth optimization enabled by the MaiaEdge platform.

The base FMO scenario assumes:

- 1.5X oversubscription enabled through traffic engineering
- 15 minutes of provisioning labor per customer site
- 10 minutes of annual network operations labor per customer site

Cloud On-Ramp TCO Results

The TCO analysis demonstrates that the MaiaEdge Future Mode of Operation (FMO) delivers substantial economic advantages compared to the traditional Present Mode of Operation (PMO) cloud on-ramp architecture. Over the five-year analysis period, the PMO generated a total TCO of approximately \$4.1 million, driven primarily by recurring Equinix port charges, cross-connects, provisioning labor, and network operations costs. The MaiaEdge FMO with 150% oversubscription reduced TCO to approximately \$1.37 million, representing a 67% reduction versus PMO. The results of the TCO analysis are presented in Figure 1: Five-year cloud on-ramp TCO comparison for the PMO and MaiaEdge FMO models. and Figure 2: Annual cloud on-ramp TCO comparison for the PMO and MaiaEdge FMO models..

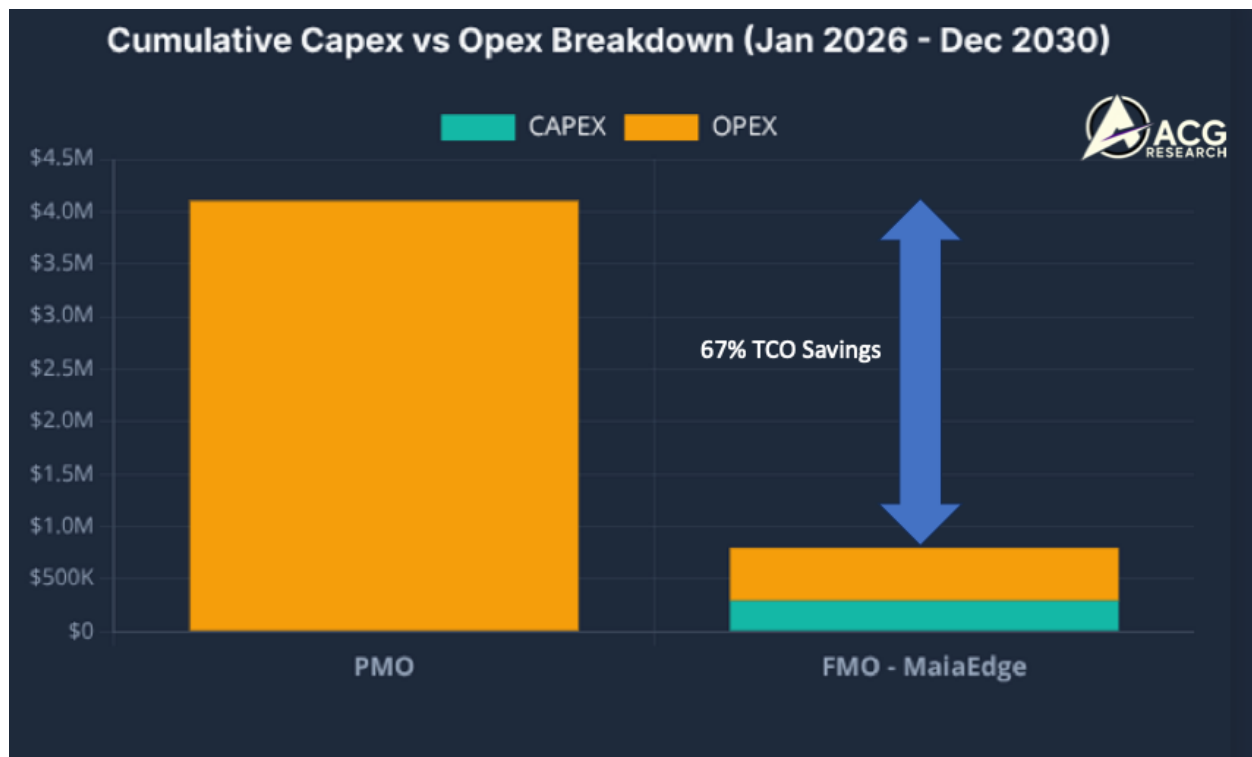


Figure 1: Five-year cloud on-ramp TCO comparison for the PMO and MaiaEdge FMO models.

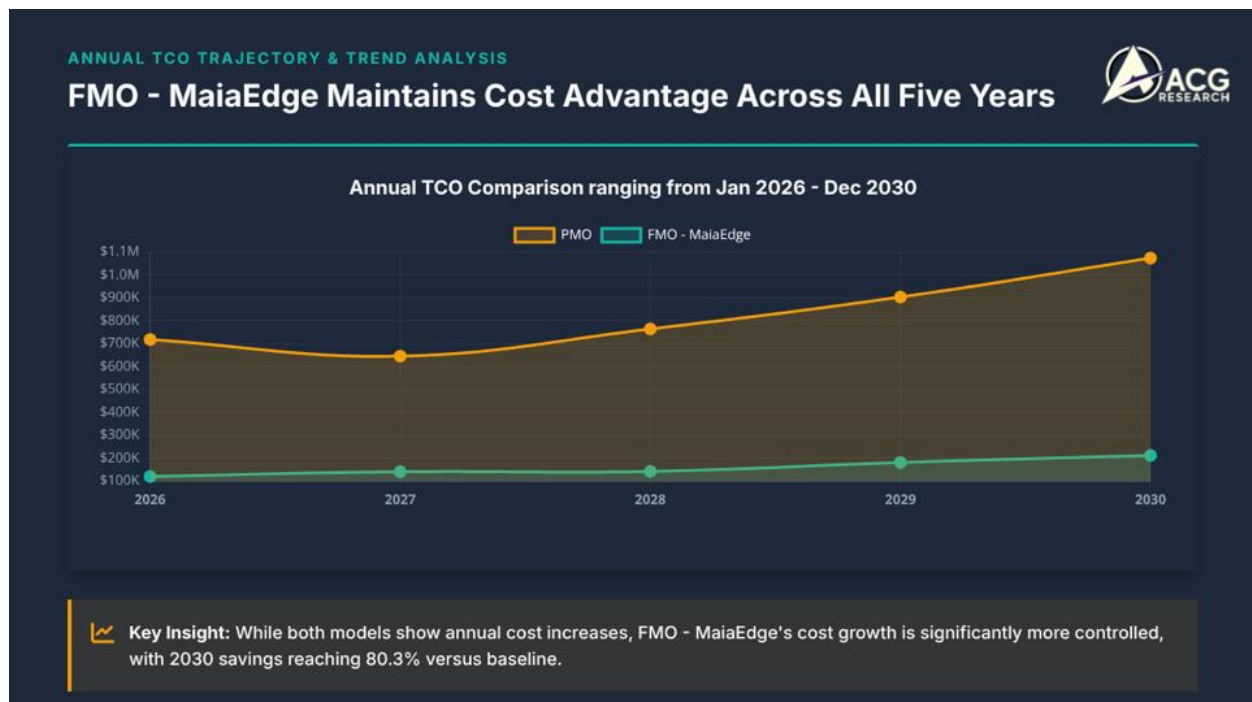


Figure 2: Annual cloud on-ramp TCO comparison for the PMO and MaiaEdge FMO models.

Figure 3: TCO savings are driven primarily by higher cloud exchange port utilization and reduced cross-connect costs. presents a breakdown of MaiaEdge cost savings in the FMO cloud on-ramp model. The largest contributor to savings was improved utilization of cloud exchange ports due to MaiaEdge traffic engineering and oversubscription capabilities. The PMO required significantly more Equinix ports and cross-connects to support customer demand due to low average utilization. MaiaEdge reduced Equinix port costs from approximately \$2.5 million in the PMO scenario to approximately \$540,000 in the FMO scenario. Cross-connect costs were also reduced from approximately \$840,000 to approximately \$180,000.

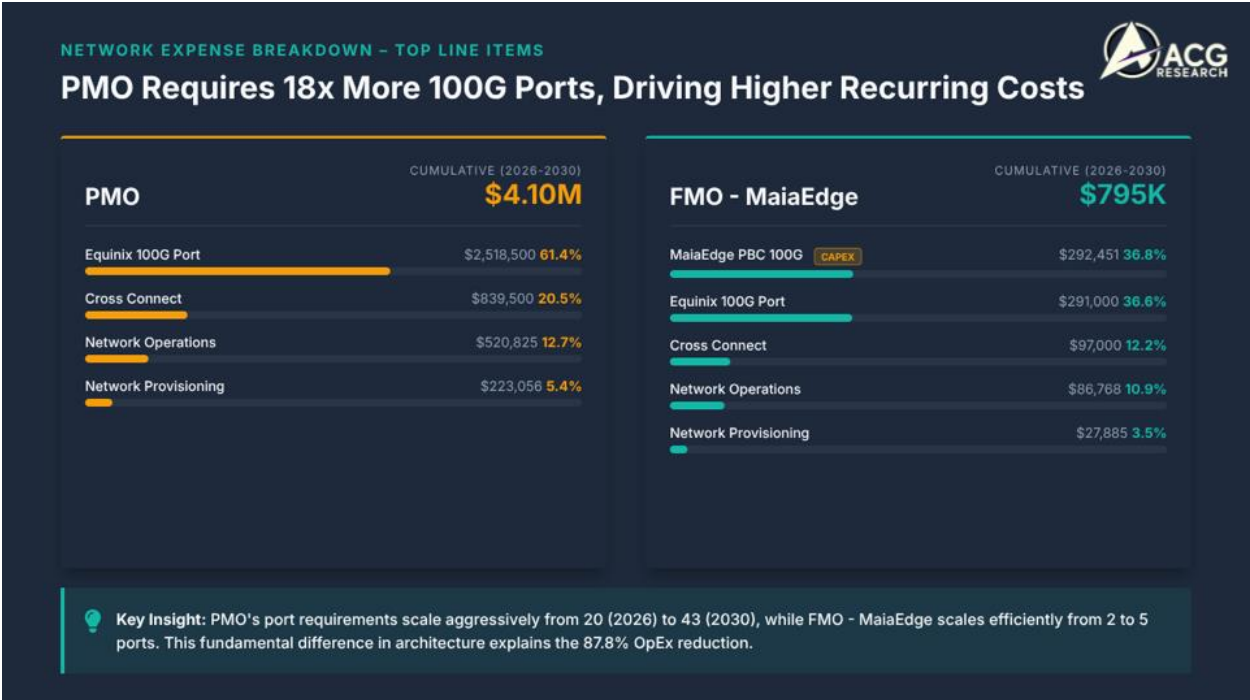


Figure 3: TCO savings are driven primarily by higher cloud exchange port utilization and reduced cross-connect costs.

Operational savings were also significant. Automated provisioning and simplified operations reduced network management and provisioning labor costs by approximately 85% compared to the PMO model.

The results show that MaiaEdge not only lowers infrastructure and operational costs but also improves scalability and enables providers to support substantially more cloud connectivity traffic using fewer physical interconnections.

Wholesale Cloud On-Ramp Business Case

In addition to reducing the cost of delivering cloud connectivity services, MaiaEdge enables service providers to create new wholesale revenue opportunities.

Many regional and Tier 3 providers do not maintain direct connectivity into major cloud exchange facilities such as Equinix or Megaport because of the cost and operational complexity of establishing cloud exchange ports and deploying dedicated infrastructure. MaiaEdge addresses this challenge by allowing providers with existing cloud exchange infrastructure to wholesale connectivity services to other operators through a marketplace model.

ACG Research modeled an initial deployment of one 100G Equinix port, with two new ports added annually as demand grows. Under the 1.5X oversubscription model, a single 100G port can support 150 Gbps of capacity. The model assumes 75% average utilization of the bandwidth which is an effective sellable bandwidth of 113G.

In the model, providers with existing cloud access sell wholesale connectivity to other providers in tiers ranging from 1G to 10G services. The model assumes an average blended wholesale selling price of approximately \$200 per Gbps per month with 3% annual price escalation.

The financial results demonstrate a strong revenue and profitability profile:

- Annual revenue grows from approximately \$270,000 in Year 1 to approximately \$2.7 million by Year 5
- Five-year cumulative cash flow exceeds approximately \$4.7 million
- EBITDA margin improves from approximately 55% in Year 1 to nearly 67% by Year 5
- EBITDA grows from approximately \$150,000 in Year 1 to approximately \$1.8 million in Year 5



Figure 4: Five-year wholesale cloud connectivity revenue and EBITDA results.

The MaiaEdge marketplace model creates a two-sided revenue opportunity: providers with cloud access can further monetize existing infrastructure, while providers without direct access can buy wholesale connectivity from partners and offer cloud services to their customers.

Ethernet over DIA vs Traditional Layer 2 Services

Dedicated private Layer 2 services can be costly or difficult to deliver in locations where a provider lacks existing Ethernet infrastructure, especially across long-haul, subsea, or international routes. MaiaEdge enables providers to deliver private Ethernet over any available transport.

ACG Research modeled a provider delivering fifty 1 Gbps private Ethernet circuits between New York and London, with 10% annual circuit growth over the five-year analysis period. The model compares the cost of delivering those services over traditional dedicated Layer 2 Ethernet circuits with the cost of delivering private Ethernet over DIA using MaiaEdge.

As shown in Figure 5, using MaiaEdge to deliver private Ethernet over DIA reduced five-year TCO by 53% compared to traditional dedicated private Layer 2 services.

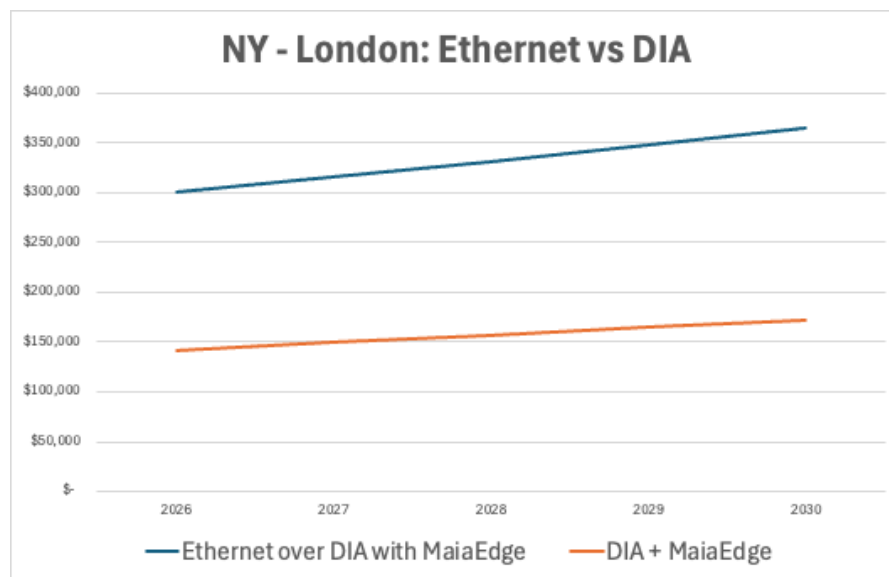


Figure 1: Ethernet over DIA with MaiaEdge reduced five-year TCO by 53% compared to traditional L2 services.

Conclusion

MaiaEdge enables service providers to simplify cloud interconnection, expand service reach, and improve profitability through automation, federation, and intelligent traffic engineering. By replacing manual provisioning and low-utilization cloud connectivity models with a software-driven architecture, providers can deliver cloud on-ramp, wholesale connectivity, and Ethernet services more efficiently and at lower cost.

ACG's analysis demonstrates that MaiaEdge can reduce cloud on-ramp TCO by more than 67%, generate profitable wholesale cloud connectivity services with margins approaching 67%, and reduce Layer 2 Ethernet delivery costs by 53% through Ethernet-over-DIA architectures. At the same time, MaiaEdge enables faster service activation, improved infrastructure utilization, and new revenue opportunities through partner ecosystems and cloud marketplace participation.

As cloud connectivity becomes increasingly strategic, MaiaEdge provides service providers with a practical path to building scalable, automated, and economically optimized interconnection services while maintaining control of the customer relationship and network experience.